

ReachNT

Remote repair triage that ranks by need, shares trips with H3 hexagons, and makes someone sign for who waits

Data analysis report

CDU IT Code Fair 2026, AI Challenge: Housing maintenance triage

Team AIC015 · Top Enders

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We acknowledge the Larrakia people, Traditional Owners of the land on which Charles Darwin University's Darwin campuses stand, and the Traditional Owners of all the Country this report discusses. This report contains no images, names or voices of people. All repair requests in it are synthetic.

Summary

Remote NT public housing has too few trades and very long distances. One emergency job in a very remote community can cost several times the same job in town, mostly in travel [6]. A schedule built to clear the most jobs per dollar keeps choosing town jobs, and nobody has to decide that remote tenants wait.

ReachNT reads free-text fault reports and ranks them by need alone; distance and cost are never inputs. It plans weekly trips around crew hours, wet-season road closures and airstrips, and uses Uber's open-source H3 hexagonal grid to let one tradesperson serve two neighbouring communities on one trip. It runs a year of requests under each setting, shows the coordinator what each costs and who waits, and asks a named person to sign one. Tenants can see why their repair waited.

On real NT geography with synthetic requests (70 communities, 8,345 houses, 37,515 requests), planning for the cheapest jobs cost \$558 per repair, but 9 in 10 urgent remote repairs took up to 62 days to fix, against 3 in town. ReachNT cost \$775 per repair (39% more), fixed 9 in 10 urgent remote repairs within 3 days, and cut the days households lived with faults by 71%. Shared trips alone saved \$68 per repair. Personal details sit in an encrypted vault on free tools hosted in Australia. We recommend DHLGCD publish its triage rules, record who accepts each equity trade-off, and co-design the tenant answer with communities before any use.

1 Introduction

1.1 The problem

The NT Government manages 5,058 remote public housing dwellings in 73 communities [1]. In December 2022, 55% of remote houses were overcrowded [2]. In 2022–23, 54% of First Nations households in very remote areas had major structural problems, against 25% in major cities [4]. Tenants report faults by phone (1800 104 076) or to a local Community Housing Officer, and the Department sorts each job into one of three categories with a response time [5]:

Table 1. NT response times for public housing repairs (DHLGCD Fact sheet FS17, version 10/25).

Category	Town	Remote
Immediate (dangerous)	4 hours	4 hours; a Remote Housing Maintenance Officer makes it safe if no contractor is available
Urgent	2 business days	5 business days
Routine	10 business days	25 business days

The published standard already gives remote houses two and a half times as long. Costs explain why: Nous Group found that emergency repairs cost 8.5 times more in very remote communities than in remote ones, and that travel can be 96% of an emergency job's cost, against 11–37% for planned work [6]. On the APY Lands, Housing SA batched about ten work orders per travel order on two-week trade rotations [7].

In 2015, tenants at Santa Teresa listed more than 600 repairs. The Tribunal (2018) and the Court of Appeal (2022) held that public housing must be at least safe, and in 2023 the High Court allowed compensation for the distress of living in unrepaired homes [11]. The ANAO found that community housing organisations could not tell tenants where their repairs were up to, because they had no access to the IT systems [10].

1.2 Purpose and scope

The brief asks for a tool that reads free-text fault reports, ranks jobs by urgency, safety and logistics, explains each job's place, makes the equity trade-off visible and leaves the decision with a person. We add one rule: urgency and logistics are separate steps. Urgency is about the tenant's need; logistics is about getting a tradesperson there. A single priority score that mixes them is how distance gets into "priority" without anyone choosing it.

The scope is responsive repairs in the 70 remote communities listed in the 2023 review of the NT remote housing partnership [2] and in the five towns their trades work from. Planned maintenance, homelands and town camps are out of scope.

1.3 Who it is for

The ReachNT portal has three views of the same record: the regional coordinator (queue, trips, the trade-off and sign-off), the tradesperson (a run sheet on their phone that works without signal), and the tenant (a repair tracker and a text message a Community Housing Officer or Aboriginal Interpreter Service interpreter can read out).

2 Methodology

2.1 Data

No public dataset of NT repair requests exists; the Menzies evaluation of Healthy Homes could not obtain job-level data either [9]. We built a real setting and filled it with synthetic requests (Table 2). Every output carries a "synthetic" label.

Table 2. Data sources. Job-level requests are synthetic; everything else is public.

Source	What we take from it
NIAA/PwC's Indigenous Consulting (2023), Review of the NPRH NT, App. B [2]	70 communities; houses estimated as overcrowded houses ÷ overcrowded share (4,481)
NTG remote communities list 2021 (CC BY)	Coordinates and land council for each community
ABS 2021 Census, Indigenous Profile I13 (CC BY 4.0)	Households renting from the housing authority in the five hub towns (3,864)
NT Annual Traffic Report 2023 restriction register (CC BY)	Wet-season closures for 14 communities, e.g. Lajamanu Road closed Nov–Apr
Road Report NT live feed; OurAirports	Current closures; airstrips within 8 km
DHLGCD FS17 [5]; RTA 1999 (NT) s 63 [16]; Healthabitat [8]	Response clocks; emergency repairs in law; order of health impact
Grealy et al. 2022, Table 3 [7]; Nous 2017 [6]	Job mix by trade (38,492 APY jobs); cost structure for calibration
Digital Earth Australia Landsat geomedian 2024; Esri World Imagery	Satellite basemap in the portal (offline and online)

Each community belongs to the nearest of five crew bases (Darwin, Katherine, Tennant Creek, Alice Springs, Nhulunbuy) by travel time. Access bands follow Nous: more than 2 hours is remote, more than 6 very remote. Ten island communities are fly-in only; 23 are cut by road in some wet-season weeks (Figure 1).

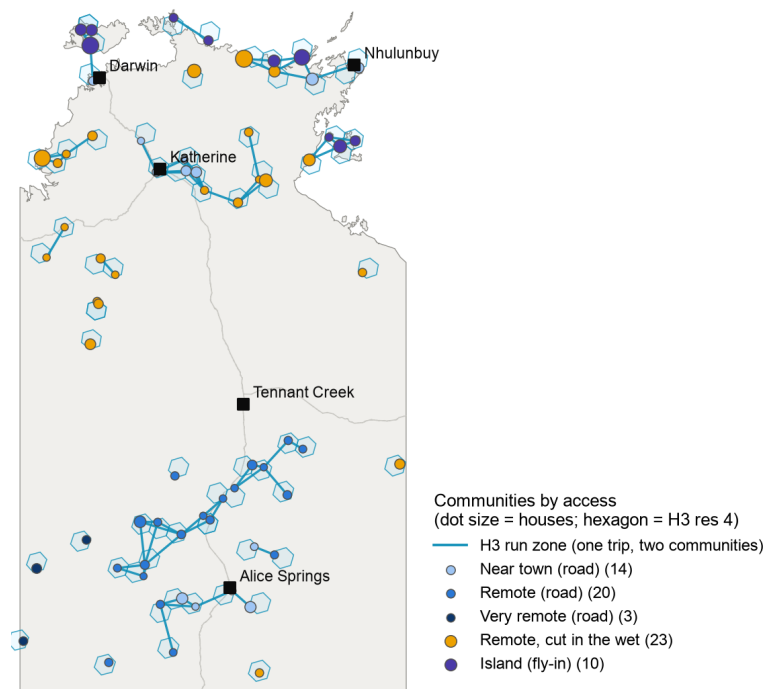


Figure 1. The 70 communities by access band, each community's H3 resolution-4 hexagon, and the 68 possible shared trips (pairs of communities within two hexagon rings that can share a trip).

2.2 Reading a fault report

Phrase rules, written from the FS17 examples and the s 63 emergency list, match 22 fault types and four modifiers (vulnerable occupant, crowded house, repeat report, negation). A TF-IDF and logistic regression model (scikit-learn) catches wording the rules miss. A person reads the report when no rule matches and the model is under 55% sure, when the two disagree on category, when the model gives 25% or more to a dangerous fault, or when a danger word (smoke, melting, sparks, sewage) appears in a reading that is not Immediate. Both the Python engine and the portal use the same rules file.

2.3 Need-only urgency

Category sets the clock. Inside a category, points order the line:

points = category (1000/500/100) + harm (0–100) + Healthy Living Practice (Safety 40 down to HLP 9 at 4) + who lives there (vulnerable +25, crowded +10) + repeat report (+10) + waiting (+3 a day past half the clock)

Harm values adapt the HHSRS severity weighting [14]. The waiting term means no job waits forever. The only inputs are the fault, what the report says and days waited; a unit test fails if distance, cost, region or community appear in the function.

2.4 H3 hexagons

H3 is Uber's open-source hierarchical grid of hexagons [21]. Every point sits in one cell at each of 16 resolutions, and each finer level is about a seventh of the area. A hexagon has six neighbours, all the same distance from its centre, so "every cell within k rings" is a fair circle in every direction; square grids such as S2 have edge and corner neighbours at two distances [21]. ReachNT uses H3 four ways:

1. **Shared trips (shared trips).** Two communities served from the same hub share one trip when their resolution-4 cells (about 1,770 km²) are within two rings, about 90 km. That gives 68 pairs, such as Ngukurr and Rittarangu, and the three Tiwi communities.

2. **While you're out there.** A tradesperson's phone lists open jobs in their trade within three resolution-5 rings of the communities on their run.
3. **A house is a hexagon.** Operational records hold a house ID and its resolution-10 cell (a 76 m hexagon), never an address.
4. **Publishing.** Anything shown outside the Department is grouped into resolution-6 cells (36 km²) and hidden below five jobs.

H3 does not know where roads are, so ReachNT pairs it with the road-restriction register and live closures for travel.

2.5 Planning trips

Each week, for each hub and trade, a constraint solver (Google OR-Tools CP-SAT) chooses which communities get a visit, alone or on a shared trip, and which jobs are done. It maximises job value minus $\lambda \times \text{cost}$, where λ is the weight on money, within each tradesperson's 40 hours, travel included. A road trip costs vehicle running plus driving time, a fly-in costs a charter for both legs, and a run zone costs hub $\rightarrow$ A $\rightarrow$ B $\rightarrow$ hub. A community with a cut road and no airstrip cannot be reached. With these costs, one job sent alone to Kintore is 92% travel, inside Nous's "up to 96%"; ten jobs on the trip bring it to 57% (Appendix D).

Settings differ in how each job is valued, in λ , and in whether shared trips are allowed:

1. **Cheapest jobs first:** every job is worth the same; $\lambda = 1$.
2. **Cheapest first + urgent guarantee:** as 1, plus a large bonus for an urgent job about to pass the equal clock (2 business days everywhere).
3. **Need first, official allowance:** need points, deadline bonus against FS17's remote clocks; $\lambda = 0.2$.
4. **Need first + urgent guarantee:** need points and the guarantee against the equal clock; $\lambda = 0.2$ or 0.5 .
5. **ReachNT:** setting 4 with H3 shared trips. Setting 1 is also run with shared trips.
6. **Need first without a guarantee:** λ from 0 to 0.8, to trace the whole curve.

2.6 Simulation, explanation and the ledger

Requests arrive at 4 per house per year, 25% more in the wet season, with faults drawn from the APY job mix; 30% use wording the reader was not trained on. One year (52 weeks plus 8 to clear the backlog) runs under each setting, with 71 tradespeople sized at 1.3 times expected job hours. The solver is single-threaded with a fixed seed, so every run gives the same numbers.

How to read our numbers. *Cost per repair* is everything spent in the year (labour, travel, charters, overnight stays) divided by the repairs done. *9 in 10 fixed within* is the number of days by which 9 out of 10 urgent repairs were done (the 90th percentile). *Fault-days* add up every day a household lived with an unfixed fault; a dangerous fault counts more than a small one.

Each week a job is not done, the planner logs the reason: road cut, no trip because of cost, crew fully booked, or the trip came but did higher-scoring jobs first. The tenant's answer is assembled from that log, the score parts and the signed decision record, so it can only state reasons that happened. A test checks each section reads at Flesch 50 or above.

2.7 Personal information and the database

ReachNT will hold names, phone numbers and addresses, so the design keeps them in one place (Appendix F). A `pii` schema holds them encrypted with pgcrypto, readable only by intake staff through a function that logs every read. The `ops` schema holds work keyed by house ID and H3 cell. Row-level security limits a tradesperson to their trips and nearby open jobs, a tenant to their own jobs, and the public to resolution-6 summaries. The decision ledger is append-only, each row hashing the one before it. Everything runs on PostgreSQL, PostGIS and h3-pg, all free and open source: on NT Government servers for \$0 in licences, or for a pilot on Neon's Sydney region, which supports all three; its free tier has 1 GB and 100 compute-hours a project [22]. A year of data is under 1 GB.

3 Findings

3.1 Left alone, efficiency pushes remote repairs to the back

Under cheapest-jobs-first, town tenants got 90% of urgent repairs within 3 days. Remote tenants waited up to 62 days, and 59% of their urgent repairs passed the 2-day equal clock (43% passed even the official 5-day remote clock). Fault-days, the danger of each fault times the days it stayed unfixed, totalled 216,000, and 770 jobs were still open after the extra eight weeks. No setting cost less per job, which is why it looks reasonable on a cost report.

3.2 The trade-off, priced

Figure 2 places every setting on two charts. ReachNT cost \$775 per repair against \$558, an extra \$218 per repair or \$8.6 million a year across the five hubs. For that, fault-days fell 71% (216,000 to 61,700) and the time to fix 9 in 10 urgent remote repairs fell from 62 days to 3. The guarantee protects urgent jobs: need ranking without it ($\lambda = 0.4$) cost \$691 per repair and left urgent remote repairs at 16 days. Ignoring cost altogether cost \$1,002 per repair for 4% fewer fault-days than ReachNT.

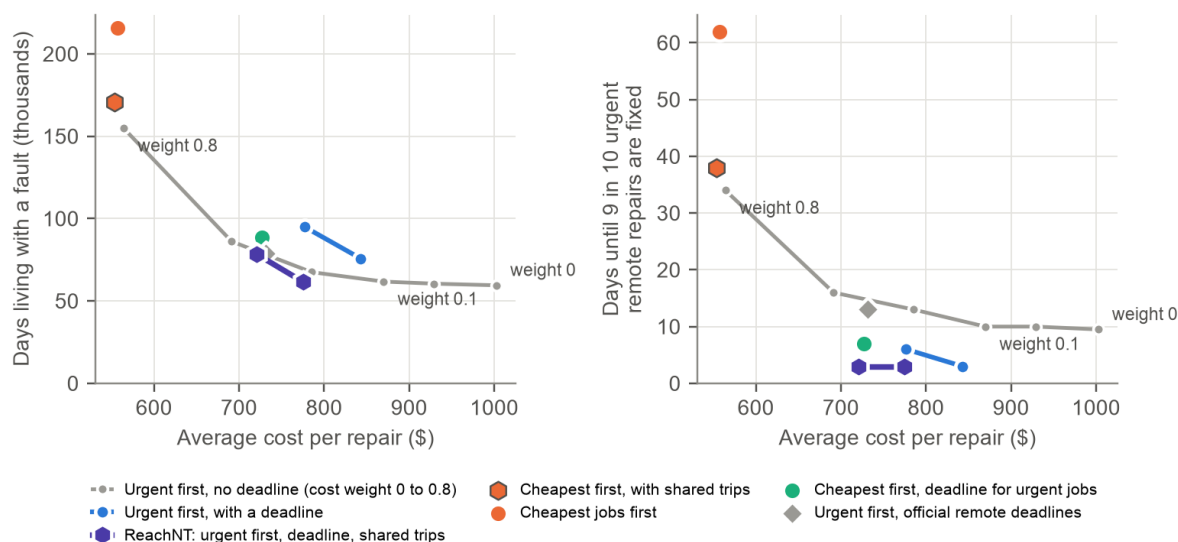


Figure 2. Average cost per repair against total fault-days (left) and against the remote urgent wait (right): days until 9 in 10 urgent remote repairs are fixed. One synthetic year, whole NT. Hexagon markers use shared trips.

3.3 What shared trips changed

Against the same setting without them, shared trips cut cost per repair from \$843 to \$775 and fault-days from 75,800 to 61,700. Communities cut in the wet went from 7 to 3 days at the 90th percentile for urgent repairs. With a tighter budget ($\lambda = 0.5$), ReachNT cost \$721 per repair, less than cheapest-first with a guarantee (\$727), with fewer fault-days (78,000 against 89,000). Even cheapest-first improves with shared trips: fault-days fell from 216,000 to 171,000 at the same cost.

3.4 Where the wait lands

Under cheapest-first, communities cut in the wet waited up to 79 days for 9 in 10 urgent repairs, and islands 71 (Figure 3). ReachNT brought these to 3 and 3. The hardest to serve are small, very remote road communities, where 90% of urgent repairs still took up to 15 days: few jobs share the cost of a long drive, and few have a neighbour within two rings.

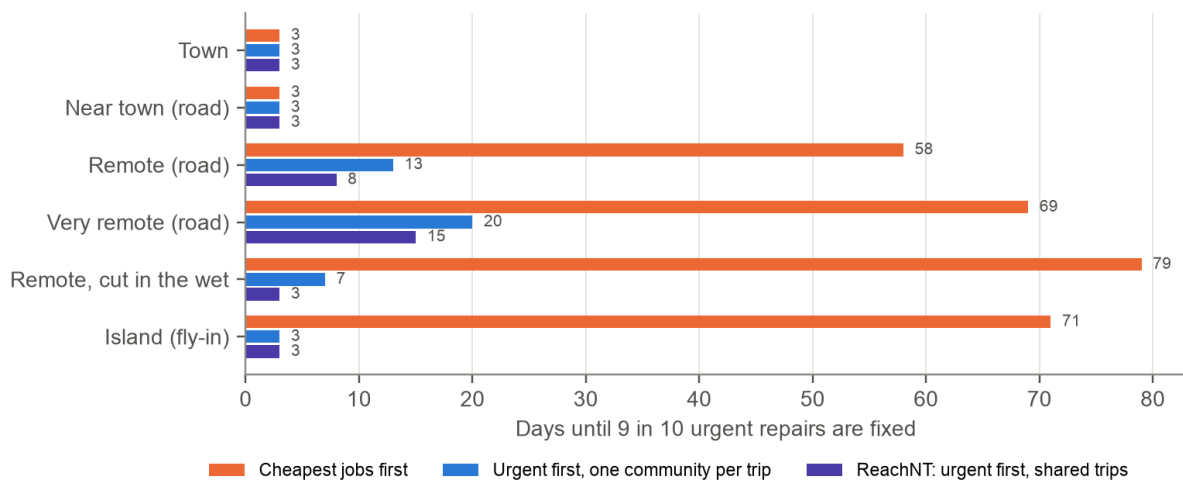


Figure 3. Days until 9 in 10 urgent repairs are fixed, by how hard the place is to reach.

3.5 Why jobs waited

The reason log (Figure 4) separates causes that look the same to a tenant. Under cheapest-first, most remote waiting was a cost decision. Under ReachNT, most of it was a fully booked crew, which calls for more trades or local trades, and a different answer to the tenant.

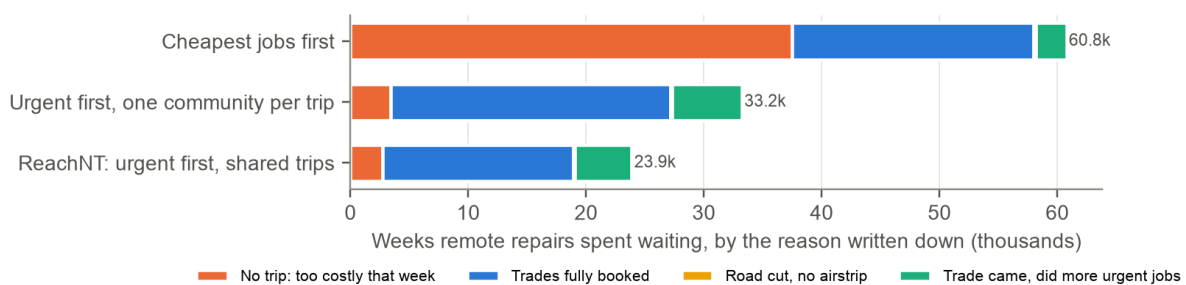


Figure 4. Job-weeks remote jobs spent waiting, by the reason the planner logged.

3.6 Reading free text is the weak link

Table 3. Reader accuracy on synthetic reports. "New wording" uses phrasings kept out of training.

Measure	Familiar wording (330)	New wording (660)
Spotting a dangerous fault, model ROC-AUC (0.5 = guessing)	1.00	0.78
Spotting a dangerous fault, model PR-AUC (guessing = 0.23)	0.99	0.60
Category correct, rules only	84%	62%
Category correct, model only	96%	59%
Category correct, combined	90%	79%
Under-triaged and not sent to a person	0.6%	3.2%
Dangerous fault missed and not sent to a person	0%	1.3%
Sent to a person	17%	40%

On new wording the model alone is only moderately good at spotting danger, and its confidence is not honest. The safety net works only because it sends four in ten reports to a person. Appendix G has the full measures: calibration, cross-validation, ranking agreement, solver gaps, and the results in four more random years. Real reports will come in Aboriginal English, Kriol and other languages [20], which synthetic English cannot test.

3.7 Sensitivity

We re-ran cheapest-first, need first with a guarantee, and ReachNT with crew capacity at 1.15 and 1.5 times job hours, the wet-season cut share at 0.2 and 0.5, the charter rate at \$1,500 and \$3,000 an hour, and a second random year (7 variants each). Cheapest-first ranged from \$541 to \$566 a job, with the remote urgent 90th percentile at 41–77 days; ReachNT ranged from \$737 to \$873 a job and 3 days in every variant. In every variant ReachNT had fewer harm-days and shorter remote urgent waits than cheapest-first. Run zones lowered both cost and harm-days in 5 of 7 variants. The direction of the trade-off holds; its size moves with the assumptions.

4 Discussion

4.1 Ethical impacts

The brief names the risk: efficiency quietly decides who waits. ReachNT answers it three ways. The urgency score cannot see place. The equity cost is shown in dollars and days before anyone chooses. The choice is signed, so a tenant told "this was a cost decision" can see who made it and when. This follows the Robodebt Royal Commission's call for plain-language information about automated decisions and a path to review, and the Ombudsman's better practice guide [17]. ReachNT recommends and a person decides. Immediate jobs are always confirmed by phone and made safe by a local officer.

The score uses what a report says about the household (a baby, an elder, a crowded house), which rewards people who say more. Intake staff should ask the same short questions every time so that it does not depend on English fluency. Privacy Act transparency rules for automated decisions start on 10 December 2026 for organisations they cover, which may include contractors [18].

4.2 Cultural and community impacts

Most remote tenants are Aboriginal, and many speak Kriol, Aboriginal English or another Aboriginal language first [20]. A reader trained on standard English will under-read them, as Table 3 shows on unfamiliar wording. ReachNT is an assistant to intake staff, never machine-translates into Aboriginal languages, and writes answers to be read out with a Community Housing Officer or interpreter.

Community-level results are sensitive: a map of long waits can read as a judgement on the community rather than the service. Under the CARE principles and Closing the Gap Priority Reform 4, communities should see the data the Department uses, first [19]. The portal uses satellite imagery of Country rather than Aboriginal art; any artwork would be commissioned and licensed through an Aboriginal art centre under the Creative Australia protocols [23]. No community has seen ReachNT. Co-design with tenants, Aboriginal Housing NT, land councils and the Aboriginal Business Enterprises that hold maintenance contracts [9] must come before any use.

4.3 Limitations

The requests are synthetic, so absolute numbers are illustrations and the comparisons between settings are the finding. Travel uses straight-line distance at a blended road speed, and wet-season cuts outside the register are an assumption. Weekly planning means even town jobs can miss a 2-business-day clock by a few days. Shared trips pair two communities; longer chains would save more. A pilot needs work-order history from the Department's TMS and ASNEX systems [9].

5 Recommendations

For DHLGCD:

1. Keep urgency and logistics as separate, published steps, and test that location is not an input to urgency.
2. Make the remote allowance in FS17 an explicit, signed decision, reviewed each quarter with its cost and its effect on remote waits, and report urgent waits by access band.
3. Plan trips on H3 shared trips, and give Community Housing Officers and tenants read access to each job's status and logged reasons, which closes the gap the ANAO found [10].
4. Build on PostgreSQL, PostGIS and h3-pg, hosted in Australia, with personal details in an encrypted vault and public outputs at resolution 6.

For contractors and industry:

5. Log a reason for every week a job waits, so cost decisions can be told apart from capacity shortfalls.
6. Price charters, batching and shared trips into contracts, and train local tradespeople in the communities that wait longest.

For communities and their organisations:

7. Co-design the tenant answer, the intake questions and the fault vocabulary in local languages with tenants, Aboriginal Housing NT and the Aboriginal Interpreter Service, under ethics approval and land council research permits.
8. Hold the decision ledger to account through local authorities and tenant groups.

References

- [1] Department of Housing, Local Government and Community Development (DHLGCD) (2024). The delivery and success of the remote housing program.
<https://dhlgcd.nt.gov.au/news/2024/the-delivery-and-success-of-the-remote-housing-program>
- [2] PwC's Indigenous Consulting for NIAA (2023). Review of the National Partnership for Remote Housing Northern Territory. <https://federalfinancialrelations.gov.au/sites/federalfinancialrelations.gov.au/files/2023-06/review-nat-partnership-remote-housing-nt.pdf>
- [3] Productivity Commission. Closing the Gap dashboard, Outcome area 9.
<https://www.pc.gov.au/closing-the-gap-data/dashboard/outcome-area/housing/>
- [4] AIHW. Indigenous Health Performance Framework, measure 2.02 Access to functional housing with utilities. <https://www.indigenoushpf.gov.au/measures/2-02-access-to-functional-housing-with-utilities>
- [5] DHLGCD (2025). Fact sheet FS17, Repairs and maintenance (version 10/25).
<https://dhlgcd.nt.gov.au/media/documents/fact-sheets/repairs-and-maintenance-fs17.pdf>
- [6] Nous Group (2017). Efficient system costs of remote Indigenous housing, report to PM&C. <https://www.niaa.gov.au/sites/default/files/documents/publications/Nous-Group-Final-Report-Efficient-system-costs-of-remote-Indigenous-housing.pdf>
- [7] Grealy, L. et al. (2022). Sustaining housing through planned maintenance in remote central Australia. Housing Studies (accepted manuscript). <https://www.hfhincubator.org/wp-content/uploads/2025/12/Grealy-et-al.-2022.-Sustaining-Housing-through-Planned-Maintenance-in-Remote-Central-Australia.pdf>
- [8] Healthabitat. Safety and the 9 Healthy Living Practices.
<https://www.healthabitat.com/what-we-do/safety-and-the-9-healthy-living-practices/>
- [9] Grealy, L., Su, J-Y. and Thomas, D. (2023). Healthy Homes Monitoring and Evaluation Project: Final Report. Menzies School of Health Research. <https://www.menzies.edu.au/content/Document/Healthy%20Homes%20Monitoring%20and%20Evaluation%20Project%20-%20Final%20Report.pdf>
- [10] Australian National Audit Office (2022). Remote Housing in the Northern Territory, Auditor-General Report No. 18 2021–22.
<https://www.anao.gov.au/work/performance-audit/remote-housing-the-northern-territory>
- [11] Grata Fund. The Ltyentye Apurte (Santa Teresa) community's fight for housing rights; High Court ruling. https://www.gratafund.org.au/santa_teresa_housing and https://www.gratafund.org.au/santa_highcourt_win
- [12] AIHW. Elective surgery waiting times by urgency category. <https://www.aihw.gov.au/getmedia/4b7b0346-84b4-435c-bc91-a9beb65413f8/myhospitals-elective-surgery-data.pdf.aspx>
- [13] UK Parliament (2025). Written statement HCWS739, Awaab's Law.
<https://questions-statements.parliament.uk/written-statements/detail/2025-06-25/hcws739>
- [14] Office of the Deputy Prime Minister (2006). Housing Health and Safety Rating System Operating Guidance. https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/15810/142631.pdf
- [15] Bertsimas, D., Farias, V. and Trichakis, N. (2011). The price of fairness. Operations Research 59(1), 17–31.
- [16] Residential Tenancies Act 1999 (NT) s 63.
https://www.austlii.edu.au/cgi-bin/viewdoc/au/legis/nt/consol_act/rta1999207/s63.html
- [17] Royal Commission into the Robodebt Scheme (2023). Report, Recommendation 17.1; Commonwealth Ombudsman (2025). Automated decision-making better practice guide.
<https://apo.org.au/node/306481>
- [18] Gilbert + Tobin (2026). Automated decision-making transparency under the Privacy Act.
<https://www.gtlaw.com.au/insights/automated-decision-making-transparency-under-the-privacy-act>
- [19] Global Indigenous Data Alliance. CARE Principles for Indigenous Data Governance.
<https://www.gida-global.org/care> ; NIAA (2024) Framework for Governance of Indigenous Data.
- [20] ABC News (2022). More than 80pc of Aboriginal people speak Kriol or Aboriginal English.
<https://www.abc.net.au/news/2022-05-05/kriol-aboriginal-english-translation-interpreting/101005782>

- [21] Brodsky, I. (2018). H3: Uber's Hexagonal Hierarchical Spatial Index. <https://www.uber.com/au/en/blog/h3/> ; H3 documentation and S2 comparison. <https://h3geo.org/docs/> ; <https://github.com/uber/h3>
- [22] Neon. Postgres extensions; regions; pricing. <https://neon.com/docs/extensions/pg-extensions> ; <https://neon.com/docs/introduction/regions> ; <https://neon.com/pricing> ; h3-pg: <https://github.com/zachasme/h3-pg>
- [23] Creative Australia. Protocols for using First Nations Cultural and Intellectual Property in the Arts. <https://creative.gov.au/first-nations-arts/protocols-for-using-first-nations-cultural-and-intellectual-property-in-the-arts>
- [24] Google OR-Tools (CP-SAT). <https://developers.google.com/optimization> ; scikit-learn. <https://scikit-learn.org> ; MapLibre GL JS. <https://maplibre.org>

Appendix A: AI usage declaration

We used Claude (Anthropic) as a coding and writing assistant: to search for and summarise public sources, draft and review Python and JavaScript, and draft and edit report text. The NT sources behind the figures we quote (DHLGCD FS17, the NIAA review, Nous 2017, Greal et al. 2022, the Menzies Healthy Homes evaluation, the ANAO audit) were read in full and the figures checked against them (copies in research/sources/). Other references, mainly overseas methods and legal context, were checked through their published summaries. All numbers in this report come from run_all.py through outputs/numbers.json. The team reviewed, tested and is responsible for all content. ReachNT itself uses no large language model: the reader is phrase rules plus a logistic regression, and tenant answers are assembled from logged facts.

Appendix B: Code, portal and how to run

```
pip install -r requirements.txt
```

```
python run_all.py rebuilds every number, figure and the portal data (about 15 minutes on 8 cores)
```

```
python run_all.py --quality quality measures only (Appendix G)
```

```
pytest -q tests checks: location-free urgency, reader safety net, planner limits and optimality, H3 shared trips, readable answers
```

```
https://reachnt.vercel.app the ReachNT portal (or open web/index.html); works offline with bundled satellite tiles
```

```
notebooks/01_walkthrough.ipynb the method step by step
```

Code: src/reachnt/ geo.py (communities, access, H3 cells and shared trips), synth.py, intake.py (reader), urgency.py (need score), planner.py (CP-SAT trips and shared trips), simulate.py, explain.py (tenant answers), experiments.py, evaluate.py (quality measures), figures.py, export_web.py. Parameters, each tagged as sourced or assumed: config/params.yaml. Fault taxonomy: config/taxonomy.yaml. Database: docs/schema.sql and docs/DATABASE.md.

Appendix C: Dataset links

NTG remote communities list:

<https://data.nt.gov.au/dataset/remote-communities-with-mobile-coverage>

ABS 2021 Census DataPacks (Indigenous Profile, ILOC):

<https://www.abs.gov.au/census/find-census-data/datapacks>

NT Annual Traffic Report 2023: <https://data.nt.gov.au/dataset/annual-traffic-report-2023>

Road Report NT: <https://roadreport.nt.gov.au>

OurAirports: <https://ourairports.com/data/>

Digital Earth Australia Landsat geomedian: <https://www.dea.ga.gov.au>

ABS ASGS 2021 boundaries (coastline): <https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3>

Appendix D: Assumptions and calibration

Table D1. Cost calibration. One 2.5-hour job sent alone, and ten jobs sharing one trip, against Nous (2017): travel up to 96% of an emergency job's cost, 11–37% for planned work.

Community	Mode	One job: total	One job: travel share	Ten jobs: travel share	Ten jobs: cost per job
Kintore	road	\$3,911	92%	57%	\$748
Lajamanu	road	\$3,488	91%	54%	\$705
Wadeye	road	\$2,377	86%	45%	\$594
Galiwinku	air	\$4,784	93%	61%	\$835
Barunga	road	\$793	59%	13%	\$372

Main assumptions (all in config/params.yaml): labour \$130 an hour; vehicle \$1.10 a km; overnight \$320; charter \$2,200 an hour; blended road speed 70 km/h with 1.2 circuitry; requests 4 per house per year; crews $1.3 \times$ expected job hours; 35% of wet-season weeks cut for Top End communities more than 2 hours out that the register does not cover.

Appendix E: A tenant's answer (synthetic)

An urgent job at the 90th percentile of waits in a community cut in the wet, under cheapest-first (J33009; community name withheld). Text message: "Your no hot water repair is urgent. Fixed after 72 days. Held up by travel cost. Ask why: 1800 104 076"

What we heard. You told us: "water is cold in the shower, lots of people staying, grandkids staying with us, can someone come". We read this as: no hot water.

How urgent it is. This is an Urgent repair. Our rule is the same in town and out bush: a plumber within 2 business days. The law calls this kind of fault an emergency repair.

Where it is up to. Fixed after 72 days. That is longer than our rule, by 69 days.

What held it up. For 21 days no plumber was sent to [community]. It costs about \$2,900 to drive one there and back. That was a cost decision, not a judgement about your repair. Nobody signed off on it. For 35 days every plumber was fully booked on repairs that scored higher than yours. Higher scores mean more danger or a longer wait. For 14 days a plumber was working nearby but did higher-scoring repairs first.

What would change it. If it gets worse (sparks, smoke, water everywhere, sewage), call 1800 104 076 and say so. It becomes Immediate and is made safe the same day. Every day you wait adds points, so your repair moves up the line.

Your rights. You can ask for an Aboriginal interpreter at any time. If an urgent repair is not done, you can ask the Tribunal (NTCAT) for an emergency repair order. You can also complain to the NT Ombudsman.

Score 838: 500 for the urgent category, 55 for harm, 36 for the health practice it affects, 35 for who lives there, 0 for a repeat report, 212 for waiting. Distance and cost are never part of this score.

Appendix F: Who sees what

Table F1. Data zones and access in the ReachNT database (docs/schema.sql).

Zone	Holds	Who can read it
pii (vault)	Name, phone, address, language, interpreter need; all personal fields encrypted	Intake staff through a logged function; the tenant through a one-time code
ops (work)	Houses as ID + H3 res-10 cell, jobs, scores, trips, weekly reasons, decision ledger	Coordinator (own hub); tradesperson (own trips + open jobs within 3 res-5 rings); housing officer (own communities); analyst (no vault)
public (views)	Waits and fault-days by H3 res-6 cell and access band, cells under 5 jobs hidden	Anyone

Appendix G: How good are the algorithms?

Every measure below is computed by `src/reachnt/evaluate.py` on synthetic reports: 330 in familiar wording and 660 in wording kept out of training, 23% of them dangerous (Immediate). ROC-AUC is the chance the model scores a random dangerous report above a random safe one (0.5 is guessing, 1 is perfect). PR-AUC summarises how many flagged reports are truly dangerous across all thresholds; guessing scores the share of dangerous reports, 0.23. Ranges are 95% bootstrap intervals.

Table G1. Reader quality. "Model" is the logistic regression alone; "system" is rules, model and the send-to-a-person checks together.

Measure	Familiar wording	New wording
Dangerous fault, ROC-AUC (model)	1.00 (0.99–1.00)	0.78 (0.73–0.82)
Dangerous fault, PR-AUC (model)	0.99 (0.98–1.00)	0.60 (0.53–0.68)
Dangerous reports caught: labelled Immediate or sent to a person (system recall)	100%	99%
Dangerous reports labelled Immediate straight away	99%	85%
Flagged reports that were truly dangerous (system precision)	56%	40%
Category (Immediate/Urgent/Routine), macro ROC-AUC (model)	0.99	0.76
Category, macro PR-AUC (model)	0.99	0.64
Category, accuracy / macro-F1 (system)	90% / 0.90	79% / 0.79
Category, quadratic-weighted kappa (system)	0.88	0.75
Exact fault type (22 types), top-1 / top-3 accuracy (model)	95% / 100%	42% / 75%
Exact fault type, macro-F1 (model)	0.95	0.39
Calibration: Brier score / expected calibration error (model, lower is better)	0.12 / 0.13	0.77 / 0.13

Five-fold cross-validation on the training reports gave macro-F1 0.92 ± 0.01 and danger ROC-AUC 0.99 ± 0.00 , so the model is stable on wording it knows. The drop on new wording is the real finding: the model alone separates dangerous from safe reports only moderately (ROC-AUC 0.78), and its confidence runs ahead of its accuracy at the top (Figure 5, right). The system still catches 99% of dangerous reports because it sends doubtful ones to a person, at the price of a precision of 40%. For that reason ReachNT never shows tenants a confidence percentage.

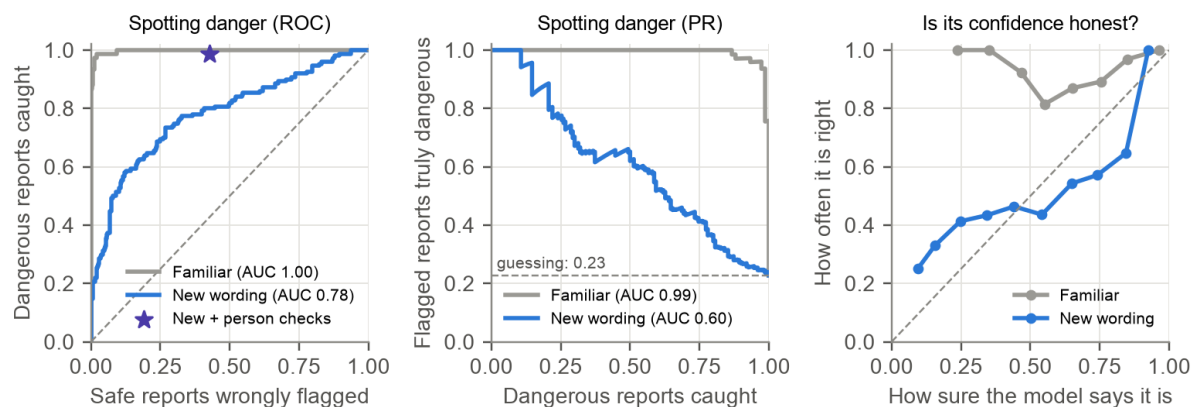


Figure 5. Left and centre: how well the model's danger score separates dangerous from safe reports; the star is the whole system on new wording, with a person checking doubtful reports. Right: when the model says it is X% sure, how often it is right (the dashed line is honest confidence).

Moving the confidence threshold from 0.30 to 0.90 changed the share sent to a person only between 38% and 40%: most referrals come from the danger checks, not the threshold. We kept 0.55, set before testing; tuning it on the test reports would flatter the result.

Ranking. On new wording, the queue built from read reports agreed with the queue built from the true faults at Kendall's $\tau = 0.65$ (0.86 once a person has corrected the reports sent to them). Of the 150 reports at the top of the queue, 77% were truly dangerous before that check and 93% after.

Planner. Across the 26,028 weekly plans in the runs below, CP-SAT proved 99.5% of them optimal: no better plan exists under the model's rules. The other 0.5% stopped at the 2-second limit with a feasible plan; for those the solver can only bound how much better a plan might be, at most 83% in the worst week, and such bounds are usually loose. The median plan took 4 milliseconds and 95% took under 0.08 seconds.

Table G2. The headline plans in five random years of synthetic requests (the main year and four more). Mean, with the range.

Plan	Cost per repair	9 in 10 urgent remote repairs fixed within (days)	Fault-days (thousands)
Cheapest first	\$561 (\$558–\$564)	61 (58–62)	206 (194–216)
Urgent first, one community per trip	\$852 (\$843–\$864)	4 (3–6)	73 (64–77)
ReachNT	\$781 (\$775–\$786)	3 (3–3)	59 (50–62)

In every year ReachNT cut fault-days against cheapest-first by 69%–74%, for 38%–40% more per repair, and shared trips saved \$68–\$78 per repair against the same plan without them.